Full Technical Specification: Unified CRF Engine

Project Scope: Multi-Source Case Report Form (CRF) Generation & Management **Core Architecture:** Unified CRF Engine with AI-Driven Protocol Analysis and Vision OCR

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1. Architectural Strategy: Unified CRF Engine

The system must support two distinct sources for CRF creation while maintaining a single, unified output schema.

Creation Paths:

1. **Proposal-Based Generation:** AI analyzes the research protocol/proposal to extract variables and visits.
2. **Existing CRF Import:** OCR and Vision processing of uploaded images or PDF forms.
3. **Hybrid Mode:** Simultaneous analysis of both proposal and existing forms with a comparison engine.

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2. Path A: Proposal-Based CRF Generation

When a researcher uploads a research proposal, the AI Protocol Analysis must extract:

- **Study Metadata:** Population, Eligibility, Interventions, Controls.
- **Outcomes:** Primary and Secondary outcomes.
- **Workflow:** Visits, Timepoints, Follow-up schedules.
- **Technical Requirements:** Imaging (Panoramic, Ceph), Measurements, STL requirements.

Output: A draft CRF automatically populated with the extracted variables.

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3. Path B: Existing CRF Import (OCR/Vision)

Support for JPG, JPEG, PNG, and PDF. The system must not just save the image; it must perform:

- **Layout Analysis:** Detecting sections and groupings.

- **Field Detection:** Identifying input areas, checkboxes, and tables.
- **Semantic Classification:** Mapping detected text to structured field types (e.g., “Age” -> Number Field).

Output: A structured CRF schema reflecting the original form’s logic and layout.

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4. Path C: Hybrid Comparison Engine (The Power Mode)

If both a proposal and an existing form are provided, the **CRF Comparison Engine** must:

- Compare **Protocol Requirements** vs. **Existing Form Fields**.
- Identify **Missing Variables** (required by protocol but missing in form).
- Identify **Extra Variables** (in form but not in protocol).
- Detect **Conflicts** (different units, data types, or definitions).
- **AI Suggestions:** Recommend additions or modifications for researcher approval.

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5. Unified CRF Schema (The Core Output)

Regardless of the source, the output must adhere to a standardized JSON schema:

```
{
  "crf_id": "string",
  "study_id": "string",
  "version": "1.0",
  "source_type": "PROPOSAL | IMPORTED | HYBRID",
  "sections": [
    {
      "id": "baseline",
      "title": "Baseline Examination",
      "fields": [
        {
          "id": "age",
          "type": "number",
          "label": "Age",
          "metadata": {
            "source": "IMPORTED_CRF",
            "traceability_id": "field_001"
          }
        }
      ]
    }
  ]
}
```

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6. Unified CRF Builder & Dental Components

A single builder interface for all study types, supporting advanced dental-specific components:

- **Standard Fields:** Text, Number, Date, Select, Checkbox, Scale.
- **Imaging Fields:** Panoramic, Cephalometric (with WebCeph integration adapter).
- **3D Components:** STL Upper Arch, STL Lower Arch, STL Occlusion.
- **Advanced Tools:** 3D Viewer with measurement tools (Point-to-Point distance in mm).

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7. Dental Imaging Integration

Panoramic & Cephalometric

- **Data Attributes:** Participant, Visit, Image File, Date, Notes, Radiology Report.
- **WebCeph Integration:** Automatically map returned measurements (SNA, SNB, ANB, FMA) to CRF variables.

STL 3D Models

- **Viewer Features:** Rotate, Pan, Zoom, Reset, Standard Views (Front, Back, Top, etc.).
- **Measurement Tool:** Point A to Point B distance calculation saved directly to the CRF field.

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8. Traceability & Versioning

Source Tracking

Every field must track its origin:

- `PROPOSAL_GENERATED`
- `IMPORTED_FROM_FORM`
- `AI_SUGGESTED` (Accepted by researcher)
- `RESEARCHER_ADDED / MODIFIED`

Versioning Logic

- **Draft:** Editable during design.
- **Approved (v1.0):** Locked for data collection.
- **Revision (v1.1):** Created for changes after study start; original data remains linked to v1.0.

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9. Security & Audit Trail

- **RBAC:** Access to CRF design vs. data entry.
- **Audit Logs:** Immutable records of who changed what field, when, and why.
- **Validation:** Server-side enforcement of mandatory fields and data types.

10. Acceptance Criteria

1. **Source Independence:** The API `GET /crf` returns the same schema regardless of how it was created.
2. **Comparison Accuracy:** The engine correctly identifies missing protocol variables in an imported form.
3. **3D Functionality:** STL measurements are accurate and saved to the database.
4. **Traceability:** Every field in the final CRF has a clear source tag.
5. **Versioning:** Changes to an active CRF trigger a new version without data loss.